Case report of the broad spectrum of late complications in an adult patient with univentricular physiology palliated by the Fontan circulation.

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At the most severe end of the spectrum of congenital heart disease are patients with an univentricular physiology. They comprise a heterogeneous group of congenital heart malformations that have the common characteristic that the cardiac morphology is not equipped for sustaining a biventricular circulation. Here, we present a case of an adult patient after Fontan palliation, illustrative of the complex clinical course and the broad spectrum of complications that can be encountered during follow-up, highlighting the need for a multidisciplinary approach in the clinical care for these patients. During the surgical Fontan procedure, the inferior vena cava is connected to the pulmonary circulation, after prior connection of the superior vena cava to the pulmonary arterial circulation. The resulting cavopulmonary connection, thus lacking a subpulmonic ventricle, provides non-pulsatile passive flow of oxygen-poor blood from the systemic venous circulation into the lungs, and the functional monoventricle pumps the oxygen-rich pulmonary venous return blood into the aorta. With an operative mortality of <5% and current 30-year survival rates up to 85%, the adult population of patients with a Fontan circulation is growing. This increase in survival is, however, inevitably accompanied by long-term complications affecting multiple organ systems, resulting in decline in cardiovascular performance. For optimal treatment, the evaluation in a multidisciplinary team is mandatory, using the specific expertise of the team members to timely detect and address late complications and to support quality of life.

Fontan's procedure in two stages.

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Since 1985, 381 patients with various cardiac malformations that share the central common feature of only one effective ventricle have been treated with some modification of Fontan's operation. Since 1989, the Fontan operation has been staged by associating the superior vena cava with the branch pulmonary arteries (hemi-Fontan) initially followed, some months later, by associating the inferior vena cava to the branch pulmonary arteries (completion Fontan). The difference in early mortality for a primary-Fontan operation (16%) compared with a completion-Fontan operation (7%) is substantial ($p < 0.05$). Since January 1991, mortality associated with the hemi-Fontan operation has been 5.5% (7 deaths in 127 patients). A systematic staged approach to Fontan's operation has been undertaken with a hemi-Fontan operation in patients who are 6 months of age and a completion Fontan operation when those patients are 12 to 18 months of age in an effort to reduce the volume load of the ventricle as early as possible, to minimize intermediate mortality from the palliated state, and to reduce the impact of rapid changes in ventricular geometry and diastolic function that can accompany either the hemi-Fontan or primary Fontan operation, but that are lethal only after a primary Fontan operation.

[Modified Fontan's operation].

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Nine-year-old female patient presented with cianosis since she was born, fatique and dyspnea when sucking. The diagnosis was univentricular heart with left ventricular morphology, transposition of the great arteries, moderate pulmonary valve stenosis and atrial septal defect. Submitted to surgical correction with superior vena cava-right pulmonary artery anastomosis, inferior vena cava anastomosis using lateral tunnel, with cardiopulmonary bypass. After surgical correction, the clinical and laboratorial (echocardiogram and cardiac catheterization) evaluation showed Fontan operation with good surgical results. Total cavopulmonary connection was proposed as a modification of the Fontan procedure that might have greater benefits than previous proposed techniques. The results demonstrate that this modification provides excellent early definitive treatment, increasing hemodynamic profile, with low morbidity and mortality, for a variety of complex congenital heart lesions.

Pulsatile blood flow in total cavopulmonary connection: a comparison between Y-shaped and T-shaped geometry.

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Single-ventricle anomaly is a hereditary heart disease that is characterized by anatomical malformations. The main consequence of this malformation is desaturated blood flow, which without proper treatment increases the risk of death. The classical treatment is based on a three-stage palliative procedure which should begin from the first few days of patient's life. The final stage is known as Fontan procedure, in which inferior vena cava is directly connected to pulmonary arteries without going through the ventricle. This connection is called total cavopulmonary connection (TCPC). After surgery, the single ventricle supplies adequate and saturated systemic blood flow to the body; however, TCPC contains low pressure and low flow pulsatility. To overcome this problem, a new method is proposed wherein pulsatile blood will be directed to the TCPC through the stenosed main pulmonary artery. In this study, through the use of Computational Fluid Dynamics, T-shaped (MRI-based) and Y-shaped (computer-generated) geometries are compared in order to determine the influence of this modification on pulsation of blood flow as well as energy loss in pulmonary arteries. The results indicate that energy loss in Y-shaped geometry is far less than T-shaped geometry, while the difference in flow pulsatility is insignificant.

CT for Assessment of Thrombosis and Pulmonary Embolism in Multiple Stages of Single-Ventricle Palliation: Challenges and Suggested Protocols.

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The total cavopulmonary connection (TCPC), or Fontan procedure, diverts systemic venous blood directly into the pulmonary arteries and is the palliative surgery of choice for patients with a wide variety of congenital heart diseases with single-ventricle physiologic characteristics. Pulmonary embolism and thrombosis are known complications and are among the major causes of morbidity and mortality in patients after TCPC. Magnetic resonance (MR) imaging is usually performed for postoperative evaluation of patients after single-ventricle repair; however, screening for thrombosis or embolism with MR imaging is not always feasible because of the emergent nature of the clinical presentation or because of artifacts from metallic devices or coils. Computed tomographic (CT) angiography is an effective method for diagnosing pulmonary embolism in children. However, because of altered hemodynamics after single-ventricle palliation, there are unique challenges in achieving optimal opacification of the pulmonary arteries and Fontan circuit that can result in nondiagnostic CT angiographic studies or erroneous image interpretation. Radiologists should be familiar with the multiple stages of single-ventricle palliation, understand the technique for performing pulmonary CT angiography at each stage, and recognize common pitfalls in obtaining and interpreting pulmonary CT angiographic images in patients who have undergone single-ventricle repair. Online supplemental material is available for this article. (©)RSNA, 2016.

Early and intermediate-term results of the extracardiac conduit total cavopulmonary connection for functional single-ventricle hearts.

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Fontan operation has evolved from atriopulmonary connection to total cavopulmonary connection (TCPC) due to its advantages in terms of hemodynamics and reduction of atrium-related complications. We analyzed the early and intermediate-term results of extracardiac conduit TCPC (EC-TCPC) procedure in patients with functional single ventricle to investigate the risk factors of surgical mortality and intermediate failure. Retrospective review of the medical records of 88 consecutive patients with functional single ventricle who underwent EC-TCPC from 2000 to 2013 was conducted. The follow-up was 100% complete, ranging from 3 months to 13 years (mean 7.0 ± 3.8 years). There were two (2.3%) hospital and 18 (20.4%) late deaths. The estimated event-free survival rates at 1 year, 5 years, and 10 years were 90.6%, 89.3%, and 77.2%, respectively. On univariate analysis, fenestration was the only risk factor for surgical mortality ($p = 0.027$). On multivariate analysis, the significant atrioventricular valve regurgitation was the only risk factor for intermediate failure ($p = 0.017$). The clinical results of EC-TCPC in patients with functional single ventricle were satisfactory. The patients who needed fenestration during operation had higher risk of surgical mortality. Significant atrioventricular valve regurgitation had negative impact on intermediate survival.

Comparison of Echocardiographically-Calculated Fontan Fenestration Gradient and Catheter-Based Measurement.

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Patients born with single ventricle anatomy typically undergo surgical palliation in three stages, culminating in the Fontan procedure. Assessment of flow across a Fontan fenestration by Doppler ultrasound theoretically allows for non-invasive estimation of the transpulmonary gradient (TPG). Our objective was to determine the relationship between Doppler-derived mean fenestration gradient (mFG) and direct catheter-based measurements of TPG in patients with fenestrated Fontans. We performed a single-center retrospective cohort study of 59 patients with fenestrated Fontans completed between 2000 and 2022. The primary outcome was catheter-based measurement of TPG and the primary predictor was mFG from echo performed within 6 months of the catheterization. Linear regression and R2 were used to determine the relationship between predictors and outcomes. Catheter-based measurements of TPG and mFG were weakly correlated ($R2 = 0.382$, $p < 0.001$); the regression coefficient was 0.550, with a standard error of 0.09 for every increase in mFG (Cath TPG = 0.55 [mFG] + 1.92). mFG had a slightly better predictive relationship with cath-derived TPG in patients with systemic left ventricles with R2 of 0.47, $p < 0.004$. mFG accounts for approximately 38% of the variance in catheter-derived TPG. Although mFG is non-invasive and intuitive, mFG in Fontan patients should be interpreted with caution and direct measurement by cardiac catheterization should be considered.

Four successful pregnancies in a woman after Fontan palliation: a case report.

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A Fontan operation is performed to provide palliation for patients with many forms of highly complex congenital heart disease that cannot support a biventricular circulation. Increasing numbers of women who have undergone these connections in childhood are now reaching their childbearing years, and some are becoming pregnant. The low flow and fixed cardiac output of a Fontan circulation poses several problems during pregnancy. We report the case of four successful pregnancies in a 31-year-old Tunisian woman with congenital tricuspid atresia after Fontan operation. Her pregnancies resulted in delivery of four healthy neonates. Her clinical status remained unchanged. This case suggests that patients after adequate Fontan palliation could complete pregnancy without long-term cardiac sequelae. Intensive care should be provided with specialists, including a neonatologist, anesthesiologist and cardiologist.

Pregnancy and delivery in patients with Fontan circulation: a report of two cases.

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Fontan operation is performed to provide palliation for patients with many forms of highly complex congenital heart disease that cannot support a biventricular circulation. Increasing numbers of women who have undergone these connections in childhood are now reaching their childbearing years, and some are becoming pregnant. The low flow and fixed cardiac output of a Fontan circulation poses a number of problems during pregnancy. Here, we report two cases of pregnancy and delivery with Fontan circulation. Case 1, who underwent Fontan procedure for congenital pulmonary atresia with intact vertical septum at age 7, delivered a male infant weighing 1073 g by cesarean section at 28(6/7) weeks due to massive genital bleeding. Case 2 underwent Fontan procedure for double inlet left ventricle and delivered by vacuum extraction a male infant weighing 2142 g, while monitoring central venous pressure at 37(5/7) weeks. The former had ascites and dose of diuretic had to be added at early pregnancy, and the latter had no adverse cardiac and obstetric events. These cases suggest that patients after adequate Fontan palliation could complete pregnancy without long-term cardiac sequelae, but might be complicated with cardiac or obstetrical events. Intensive care should be required with specialists, including a neonatologist, anesthesiologist and cardiologist. We have added a literature review of pregnancy with Fontan circulation, referring to previous reports.

Long-Term Effects of Percutaneous Fenestration Following the Fontan Procedure in Adult Patients with Congenital Univentricular Heart.

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BACKGROUND The Fontan procedure, performed for univentricular heart, may also include the technique of percutaneous fenestration to create a small atrial septal defect (ASD) and a right-to-left shunt. The aim of this study was to evaluate the long-term effects of fenestration in adult patients who had a Fontan procedure for univentricular heart. MATERIAL AND METHODS Fontan surgery was performed in 39 patients, including 19 (49%) patients with fenestration (Group I), and 20 (51%) patients without the fenestration procedure (Group II). Laboratory tests in both groups included echocardiography, plethysmography, cardiopulmonary exercise testing, and 24-hour Holter monitoring. RESULTS Compared with patients in Group I, patients in Group II had a significantly increased level of N-terminal pro-brain natriuretic peptide (NT-proBNP) ($p=0.04$), alkaline phosphatase (ALP) ($p=0.01$) and a significant increase in frequency of atrial fibrillation ($p=0.04$). Patients in Group I had a significantly increased systemic ventricular ejection fraction (SVEF) ($p=0.05$) and increased heart rate (HR) ($p=0.006$), heart rate reserve (HRR) ($p=0.02$), ventilatory equivalent (VE) ($p=0.01$), and VO2 peak ($p=0.05$) on cardiopulmonary exercise testing (CPET). Renal, hematologic, and ventilatory parameters, and incidence of thromboembolism showed no significant differences between the groups. CONCLUSIONS Long-term follow-up of patients who underwent Fontan procedures with percutaneous fenestration had improved single ventricular function, lower NT-proBNP levels, improved exercise capacity, and reduced ALP levels. These findings indicate that percutaneous fenestration closure should be considered for adult patients who have undergone Fontan procedure for univentricular heart.